Parallax - Copy Detection on Ingested Data

By

Ernest Baker

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Abstract

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Advisors: A.J. Scarlato

Department: Mathematics and Computer Science

Existing launch and spaceflight tracking applications ingest data through a variety of sources from governmental to private entities. This data, ingested and shared amongst each other, has no verifiable way to safeguard against data malformation originating from a corrupt input or data manipulated with malicious intent, potentially leading to a cascading data poisoning due to shared information. Our application seeks to resolve this by evaluating each piece of data against multiple data sources without assuming independence. Tamper-evident logging protects the integrity of each record and what each source asserted. Our verification system should be able to distinguish from information that is inaccurate instead of merely just stale. By purposefully injecting malformed data, we will be able to measure detection rate and false positives through the copy detection mechanisms. As the goal is to provide accurate data, provenance is an important aspect of every data source. By logging what data came from where and when, we can establish our own provenance of a source over time, therefore

weighting sources differently enhancing our system. The goal is to provide a verified view that has been cross-checked and references other sources with a quantifiable copy detection rate.

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References

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Biographical Sketch

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